Spatiotemporal Variation of Internal Solitary Waves in the Northern Bay of Biscay Based on Sequential SAR Images

Xue Yang, Zongze Li, Yawei Zhao, and Jinsong Chong, *Senior Member, IEEE*

***Abstract*— In the Bay of Biscay, internal solitary waves (ISWs) are frequently generated by the interaction between surface tides and the steep continental shelf topography. Synthetic aperture radar (SAR) is a powerful tool for observing ISWs due to its all-weather, wide-area coverage and high resolution. However, most existing studies focus on the spatial distribution and morphological features of ISWs, with limited investigation into their spatiotemporal evolution. This study examines multiple ISW groups in the northern Bay of Biscay using time-series SAR images. We analyze their evolution by deriving their wavelength and direction from the spatial-spectral distribution and motion data of SAR images. The phase velocity is estimated by tracking surface features across sequential SAR images. Additionally, we propose a novel method to invert ISW amplitude using motion constraints, combining phase velocity and peak-to-peak distance. This approach successfully characterizes ISW dynamics at various life stages, revealing their spatiotemporal variation during propagation. By integrating synchronized tidal data, we identify a clear phase-locking relationship between ISW generation and tidal currents at the continental shelf break. A comparison with in situ data confirms that the SAR-derived ISW dynamics align with traditional measurements, demonstrating the effectiveness of SAR-based ISW studies.**

***Index Terms*— Bay of Biscay, internal solitary waves (ISWs), spatiotemporal variation, time-series synthetic aperture radar (SAR) images.**

I. Introduction

THE Bay of Biscay, located in the eastern North Atlantic Ocean between the west coast of France and the northern coast of Spain, is a triangular bay with an average depth of 1715 m, a maximum depth of 5120 m, and a total area of 194 000 km^2. The continental shelf occupies approximately half of the bay, with the remaining area consisting of steep continental slopes and deep-sea plains. Semi-diurnal tides interact with the complex topography, generating internal solitary waves (ISWs) [1], [2]. These waves propagate across the deep sea and continental shelf, often referred to as internal tides due to their alignment with the tidal cycle [3]. The formation and movement of ISWs play a significant role in nutrient transport and the distribution of phytoplankton [4], [5], [6].

Early research on ISWs in the Bay of Biscay mainly relied on in situ data, such as acoustic Doppler current profilers (ADCPs) [2], [7], [8] and moored instruments [1], [9], [10]. While these observations provide detailed profiles of current velocity, temperature, and salinity, they are limited by high costs, complexity, and limited coverage, making it difficult to fully capture the temporal and spatial distribution of ISWs. The regional simulation system, as a supplement to field observations, helps reveal the relationship between large-scale physical processes and observational data. Altimeters provide a reliable data source for regional simulation systems, capable of estimating sea surface height variations caused by ISWs and meeting the detection requirements of ISWs at kilometer-scale resolution [11]. Synthetic aperture radar (SAR) provides an effective tool for ISW observation due to its all-weather capability, high resolution, and wide-area coverage. ISWs modulate sea surface currents, altering sea surface roughness and creating distinctive bright and dark stripes in SAR images [12].

Previous studies using SAR images have analyzed ISW characteristics in the Bay of Biscay. Azevedo et al. [13] examined ISW distribution in the northern and central Bay of Biscay, revealing their impact on phytoplankton abundance. New and Silva [14] analyzed the wavelengths and coherence lengths of approximately 25 groups of mode-1 ISWs in ERS SAR images. Their results indicate that the average wavelength of the ISWs is approximately 1.35 km, with an average coherence length of 21.55 km. Azevedo et al. [15] identified a key source of ISWs near the northwest coast of Spain. However, these studies primarily focused on spatial distribution and morphology, lacking a comprehensive analysis of the dynamic evolution of ISWs throughout their lifecycle.

The complex topography and tidal currents in the Bay of Biscay cause ISWs to undergo significant morphological changes and energy dissipation during propagation. Dynamic parameters such as phase velocity, wavelength, wave direction, and amplitude are essential for characterizing these changes [16], [17]. Recent advances have been made in

Received 14 January 2025; revised 4 March 2025; accepted 10 March 2025. Date of publication 14 March 2025; date of current version 28 March 2025. This work was supported by the National Natural Science Foundation of China under Grant 62231024. *(Corresponding author: Jinsong Chong.)*

Xue Yang, Zongze Li, and Jinsong Chong are with the National Key Laboratory of Microwave Imaging, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical, and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: yangxue19@mails.ucas.ac.cn lizongze20@mails.ucas.ac.cn; lily@mail.ie.ac.cn).

Yawei Zhao is with the National Key Laboratory of Microwave Imaging, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: zhaoyw@aircas.ac.cn).

Digital Object Identifier 10.1109/TGRS.2025.3551085

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

inverting dynamic parameters from SAR images [18], [19], [20], [21], [22], [23], [24], [25]. Liu et al. [26] estimated ISW phase velocity using time-series ENVISAT and ERS-2 SAR images, while Furtney et al. [27] developed an automatic extraction method for phase velocity from COSMO-SkyMed SAR images. Santos-Ferreira et al. [28], [29] demonstrated the effectiveness of SAR altimeter (SRAL) in detecting ISWs and proposed corresponding detection methods. Compared to traditional altimeters, SRAL offers significantly improved spatial resolution [30], making it highly promising for the inversion of ISW dynamics parameters. However, inverting ISW amplitude remains a significant challenge due to its complexity.

Zheng et al. [31] proposed a method for inverting ISW amplitude using single SAR images, based on the traditional Korteweg-de Vries (KdV) equation and the Alpers radar imaging model. Pan et al. [32] established a theoretical model linking radar backscatter to ISW parameters. However, these methods rely on the traditional KdV equation, which does not accurately capture the nonlinear characteristics of ISWs on the continental shelf. To address this, Xue et al. [33] proposed an iterative numerical method based on the higher-order KdV equation. While this method is less sensitive to wind fields, its accuracy depends on the quality of SAR images, and poor image quality can compromise results. Additionally, Romeiser and Graber [34] improved amplitude inversion accuracy using along-track interferometric SAR data, but its reliance on M4S software limits broader application. Jia et al. [35] combined time-series SAR images with higher-order KdV equations, but their method can produce unsolvable equations due to the high sensitivity of amplitude to phase velocity changes.

Despite progress, challenges in accuracy and robustness remain in ISW dynamic parameter inversion. A more reliable inversion method is essential for a comprehensive analysis of ISW propagation and evolution in the Bay of Biscay. This study uses time-series SAR images to analyze ISW dynamics in the northern Bay of Biscay, focusing on the inversion of key parameters—wavelength, wave direction, phase velocity, and amplitude—and exploring the spatiotemporal variations and environmental impacts of ISWs. This article is structured as follows. Section II introduces the study area and data; Section III details the inversion method for ISW dynamic parameters; Section IV presents the analysis of the spatiotemporal variation of ISWs in the Bay of Biscay; Section V summarizes the main conclusions.

II. Study Area and Data

A. Study Area

The study area is located in the northeastern Bay of Biscay, as indicated by the blue dashed box in Fig. 1. It spans the steep continental shelf in the southwest and the broader continental shelf in the northeast. Semi-diurnal tides dominate the Bay of Biscay, and their interaction with the complex topography generates ISWs [1], [2]. Numerical models and in situ observations indicate that most ISWs are formed near the −200 m isobath, propagating both toward the deep sea and across the continental shelf, with an approximate horizontal speed of 0.6 m/s on the shelf [2]. Previous studies have revealed the presence of ISW packets with multiple propagation directions near the continental shelf in the northern Bay of Biscay, suggesting their origins from various locations along the shelf break [4]. This study used time-series SAR images to capture multiple groups of ISWs generated at the continental shelf fracture zone and propagating toward the continental shelf, and a detailed analysis of their spatiotemporal variation characteristics will be conducted.

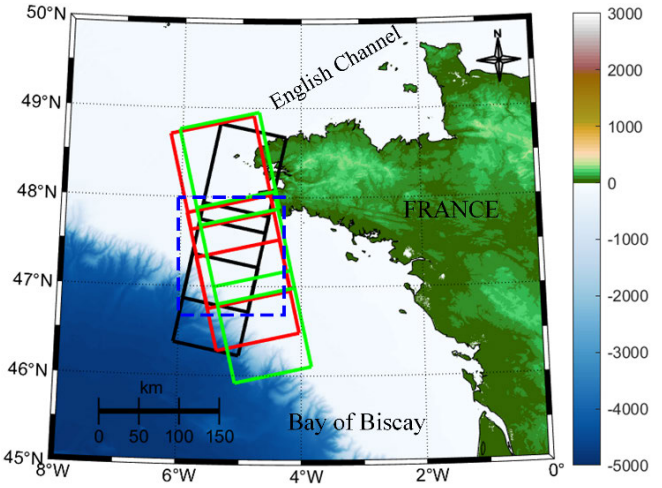


Fig. 1. Map of the Study area (blue frame). The black and red frames represent the ENVISAT SAR images acquired on June 5, 2008, at 10:37:54 UTC and 21:57:21 UTC, respectively, with a time difference of approximately 11 h and 20 min. The green frame represents the ERS-2 SAR images acquired at 22:27:38 UTC, taken approximately 30 min after the ENVISAT images (red frame).

B. Data

The ERS-2 and ENVISAT satellites, launched by the European Space Agency in 1995 and 2002, respectively, are equipped with SAR sensors that can acquire SAR images in various modes. These satellites have similar orbits, with ERS-2 trailing ENVISAT by approximately 30 min, and therefore observe nearly the same surface, potentially resulting in strong correlations between their acquired images, particularly over slowly evolving ocean features such as ISWs. The time-series SAR images used in this study were obtained in strip mode from the ESA website (https://esards.eo.esa.int/), with a pixel resolution of 12.5 m and a width of 100 km. Fig. 1 shows the coverage of the time-series SAR images. The black and red frames represent three ENVISAT SAR images acquired on June 5, 2008, at 10:37:54 UTC and 21:57:21 UTC, respectively, with an interval of about 11 h and 20 min. The green frames represent three ERS-2 SAR images acquired on June 5, 2008, at 22:27:38 UTC, approximately 30 min after the ENVISAT images (red box). Detailed information of sequential SAR images is provided in Table I.

III. Methodology

To analyze the spatiotemporal variations of ISWs in the Bay of Biscay, this study extracts ISW profiles at various stages of their life cycle and inverts their dynamic parameters—wavelength, wave direction, phase velocity, and amplitude.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DETAILED INFORMATION OF SEQUENTIAL SAR IMAGES

Sensor	Acquisition time (UTC)	Resolution (m)	Swath (km)	Polarization
Envisat ASAR	20080605 10:37:54	12.5	100	HH
Envisat ASAR	20080605 21:57:21	12.5	100	VV
ERS-2 SAR	20080605 22:27:38	12.5	100	VV

By analyzing the variations in these parameters, the structural changes and intensity evolution of ISWs throughout their life cycle are examined.

Before performing the dynamic parameter inversion, a series of preprocessing steps are applied to the SAR images, including radiometric calibration, filtering, image registration, image mosaicking, and region cropping. Radiometric calibration converts the raw intensity values into backscatter coefficients, correcting for sensor-induced radiometric biases [36], [37]. Speckle noise is reduced using a local adaptive filtering method to improve image quality. To ensure spatial consistency between SAR images acquired at different times, geographic coding and resampling are first applied, followed by region-based registration to achieve an accuracy within one pixel. Finally, SAR images from the same track are mosaicked, and the region of interest is cropped. The preprocessed sequential SAR images are then used for dynamic parameter inversion.

For the inversion of dynamic parameters from a single ISW profile, the wavelength and wave direction are first determined by applying a Fourier transform to the SAR sub-image, extracting the spatial spectrum of ISW energy. To resolve the 180° ambiguity in wave direction, corrections are made by incorporating motion information from the time-series SAR images. The phase velocity is estimated by tracking the movement of ISW features across successive images, while the amplitude is inverted using a motion-constrained method. This method combines phase velocity and peak-to-peak distance, enabling precise amplitude inversion by fitting the theoretical model to the SAR observation profile. A detailed explanation of the inversion process is provided below.

A. Wavelength and Wave Direction Estimation

Sub-image blocks with clear ISW features are selected from the preprocessed SAR images. The block size is typically 4–8 times the ISW wavelength, which yields better wavelength inversion results [38]. A 2-D FFT is applied to the sub-image blocks to obtain the spatial spectral distribution of the ISW energy. The wavelength is determined based on the peak positions, and the direction of the peak line is parallel to the propagation direction of the ISWs [39]. As a result, two potential propagation directions are identified, differing by 180°. Temporal SAR images provide motion information, which helps determine the correct propagation direction, resolving the 180° ambiguity.

B. Phase Velocity Extraction

ISWs modulate sea surface currents, amplifying (reducing) the amplitude in convergence (divergence) zones, which in turn increases (decreases) sea surface roughness, appearing as brighter (darker) stripes in SAR images [40]. Phase velocity is extracted by tracking surface feature movement across images using the maximum cross correlation method [27], [41].

First, the initial SAR image is divided into overlapping template windows. A template window T is selected in the study region and matched to the search area in the second SAR image acquired along the same track at a later time. The cross correlation coefficient r is computed as

$$r(p,q)=\frac{\operatorname{cov}\left[T(x_{\mathrm{t}},y_{\mathrm{t}}),S'(x_{\mathrm{t}}+p,y_{\mathrm{t}}+q)\right]}{\left(\operatorname{var}\left[T(x_{\mathrm{t}},y_{\mathrm{t}})\right]\operatorname{var}\left[S'(x_{\mathrm{t}}+p,y_{\mathrm{t}}+q)\right]\right)^{1/2}} \tag{1}$$

where $(x_t,\ y_t)$ is the central pixel of the template window, S' represents the corresponding search sub-window in the second SAR image, and p and q are the displacements of the search sub-window center relative to the search area center in the range and azimuth directions, respectively.

Obtain the maximum value $r_{\max}$ in the correlation coefficient matrix corresponding to the template window centered at the central pixel coordinate $(x_t,\ y_t)$, and calculate the ISW phase velocity based on the position offset of $r_{\max}$ relative to the center point. The calculation of the ISW phase velocity is as follows:

$$V=\sqrt{(p_{\max}\Delta x)^2+(q_{\max}\Delta y)^2}\Big/\Delta t \tag{2}$$

where Δt is the time interval between consecutive SAR images, Δx and Δy are the pixel intervals in the range and azimuth directions, and $p_{\max}$ and $q_{\max}$ are the number of pixels moved to the right and downward in the search area, respectively, when the template window T achieves the maximum cross correlation coefficient.

The process is repeated for the next template window in the ISW region, performing phase velocity estimation until all template windows in the ISW region of interest in the first SAR image are processed.

The accuracy of ISW phase velocity extraction depends on the time interval between sequential SAR images being smaller than the ISW decorrelation time. When Δt is small, the ISW features in the time-series SAR images are well correlated, leading to accurate phase velocity extraction. As Δt increases, some features begin to decorrelate, but most areas remain correlated, so the extraction results are still accurate. When Δt becomes large, the features decorrelate completely, making phase velocity extraction impossible. Studies show that a 30-min interval between SAR images provides high accuracy for phase velocity extraction [26], [42].

The template window size also affects accuracy. It must be large enough to capture sufficient features to track the ISW structure but not too large to avoid excessive computational

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

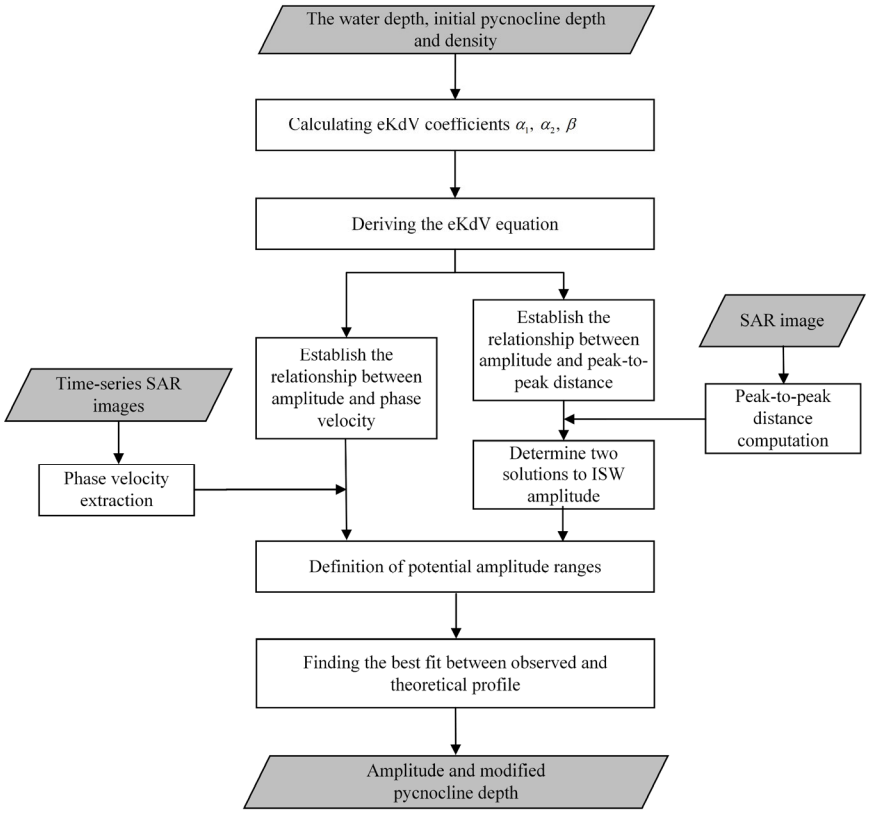


Fig. 2. Flowchart of the ISW amplitude inversion method constrained by motion information.

cost. Additionally, window overlap impacts the spatial resolution of extracted velocity vectors. Too much overlap increases computation, while too little causes inconsistencies. Therefore, the window size and overlap should be adjusted based on specific requirements.

To ensure the reliability of the extracted ISW phase velocities, the raw velocity field obtained from the MCC method must be filtered. The correlation coefficient filter and the reciprocal filter are commonly used to eliminate erroneous vectors. First, a minimum correlation threshold is applied to remove vectors with low cross correlation values, effectively discarding most unreliable vectors while some outliers may remain. The reciprocal filter [43] allows for a lower correlation threshold, increasing the number of valid vectors while eliminating obviously incorrect ones. After applying the reciprocal filter, anomalous vectors are removed, yielding the ISW phase velocity field.

In cases where the ISW motion direction deviates from the ISW normal due to interference from other ocean phenomena, the velocity component along the ISW normal is taken as the ISW phase velocity to minimize the impact of these deviations on phase velocity estimation.

C. Amplitude Inversion Constrained by Motion Information

To inverse the ISW amplitude, this article presents a ISW amplitude inversion method constrained by motion information, as outlined in Fig. 2. This method integrates the ISW phase velocity and peak-to-peak distance to estimate the amplitude. The process for extracting the ISW phase velocity has been described previously. The following steps are involved: 1) calculation of peak-to-peak distance, 2) definition of potential amplitude ranges; and 3) optimization and determination of the best-fit amplitude.

1) Calculation of Peak-to-Peak Distance: The peak-to-peak distance of ISWs represents the horizontal separation between the bright and dark stripes in SAR images and is crucial for SAR ISW amplitude inversion. In each SAR image, a profile is selected that intersects the ISW and is perpendicular to its crest line. This profile provides the horizontal distribution of radar backscatter coefficients within the ISW region. The peak-to-peak distance is then calculated by measuring the horizontal distance between the positions of the positive and negative wave crests.

2) Definition of Potential Amplitude Ranges: The inversion of ISW amplitudes from SAR images is primarily based on the classical KdV equation [31], which describes shallow water ISWs in a two-layer stratified ocean. Due to its analytical solution, the KdV equation is widely used to derive the fundamental dynamic characteristics of ISWs. However, it has limitations in capturing extreme nonlinear phenomena encountered during ISW propagation on the continental shelf. To address this, Stanton and Ostrovsky [44] proposed a higher-order KdV equation to better model the extreme nonlinear behavior of ISWs. The equation is expressed as

$$\frac{\partial \eta}{\partial t} + \left(c_0 + \alpha\eta + \alpha_1\eta^2\right)\frac{\partial \eta}{\partial x} + \beta\frac{\partial^3 \eta}{\partial x^3} = 0 \tag{3}$$

where $\eta(x, y)$ represents the vertical displacement of the fluid interface relative to the equilibrium position; c_0 is the linear

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

phase velocity, and α, $\alpha 1$, and β are the coefficients of the equation. In the two-layer fluid model, these parameters are expressed as follows:

$$c_0 = \left(\frac{g\Delta\rho h_1 h_2}{\rho_0 (h_1 + h_2)} \right)^{\frac{1}{2}} \tag{4}$$

$$\alpha = \frac{3}{2} c_0 \frac{h_1 - h_2}{h_1 h_2} \tag{5}$$

$$\alpha_1 = \frac{3c_0}{(h_1 h_2)^2} \left[\frac{7}{8} (h_1 - h_2)^2 - \left(\frac{h_2^3 + h_1^3}{h_1 + h_2} \right) \right] \tag{6}$$

$$\beta = \frac{c_0}{6} h_1 h_2 \tag{7}$$

where h_1 and h_2 represent the depths of the upper and lower water layers, respectively, $\Delta\rho$ is the density difference between the two layers ($\Delta\rho = \rho_2 - \rho_1$), and ρ_0 is the average density, given by $\rho_0 = (\rho_2 + \rho_1)/2$.

The ISW solution to the high-order KdV equation can be expressed in various forms, one of which is

$$\eta = \frac{\eta_0}{b + (1 - b)\cosh^2 \gamma (x - Vt)} \tag{8}$$

where t represents time, and intermediate parameters are defined as follows:

$$V = c_0 + \frac{\eta_0}{3} \left(\alpha + \frac{1}{2} \alpha_1 \eta_0 \right) \tag{9}$$

$$\gamma^2 = \frac{\eta_0 \left(\alpha + \frac{1}{2} \alpha_1 \eta_0 \right)}{12\beta} \tag{10}$$

$$b = \frac{-\eta_0 \alpha_1}{2\alpha + \alpha_1 \eta_0}. \tag{11}$$

In order to establish the relationship between the ISW amplitude and peak-to-peak distance, we calculate the modulation effect of ISWs on the radar backscatter. The ISW solution is used to represent the ISW structure, followed by deriving the horizontal current field induced by the ISW. The expression for the horizontal current field is

$$u = c_0 \eta(\xi) \frac{d\phi}{dz}, \quad \xi = x - Vt \tag{12}$$

where $\phi(z)$ is the first mode vertical structure function.

The modulation effect on the radar backscatter coefficient is mainly due to the redistribution of surface capillary-gravity waves, driven by the ISW-induced current field gradient, which impacts the backscatter intensity. The modulation is expressed as [35]

$$\begin{aligned} \sigma_0^{\mathrm{ISW}} &= 8\pi k_0^4 \left| g_{ij}(\theta) \right|^2 m_3^{-1} \cos\varphi k^{-4} \omega^{-1} \left(-\frac{\partial (u|z=0)}{\partial x} \right) \\ &= Q \frac{\sinh(\gamma\xi)\cosh(\gamma\xi)}{\left(b + (1-b)\cosh^2(\gamma\xi) \right)^2} \end{aligned} \tag{13}$$

where

$$Q = 16\pi k_0^4 \left| g_{ij}(\theta) \right|^2 \omega^{-1} m_3^{-1} \cos\varphi k^{-4} c_0 \eta_0 \gamma (1 - b) \left(\frac{d\phi}{dz}\bigg|_{z=0} \right). \tag{14}$$

k_0 represents the radar wavenumber, θ is the incident angle, and k is the Bragg wavenumber, i and j represent the polarization modes of the incident and reflected waves, $g_{ij}(\theta)$ denotes the radar backscatter coefficient, ω is the Bragg angular frequency, m_3 is a constant, and φ is the viewing angle of the radar beam relative to the wave direction.

The peak position of the radar backscatter coefficient induced by ISWs shifts from its equilibrium position and satisfies

$$\frac{\partial \sigma_0^{\mathrm{ISW}}}{\partial x} = 0. \tag{15}$$

By solving (14), the peak-to-peak distance of the ISW is given by [35]

$$d_{\text{p-p}} = \xi_{\max} - \xi_{\min} = \frac{2}{\gamma} ar\cosh \sqrt{\frac{(b-3) - \sqrt{9b^2 - 14b + 9}}{4(b-1)}} \tag{16}$$

where $\xi_{\max}$ and $\xi_{\min}$ represent the positions of the positive and negative peaks of the ISW, respectively.

Using (9), the ISW amplitude as a function of phase velocity is plotted, while (15) is employed to plot the ISW amplitude as a function of peak-to-peak distance. Based on the peak-to-peak distance $d_{\text{p-p}}$ obtained from the SAR image profile, two possible ISW amplitude solutions are derived from the amplitude versus peak-to-peak distance curve. The unique ISW amplitude solution is then determined by comparing it with the amplitude versus phase velocity curve.

The unique amplitude solution serves as the center, and preset ranges on either side are selected as potential intervals for the ISW amplitude.

3) Optimization and Determination of the Best-Fit Amplitude: The ISW amplitude is varied within its possible range, while the upper layer depth is adjusted within the determined variation range, centered around the estimated upper layer depth. The optimal ISW amplitude and upper layer depth are determined by minimizing the difference between the observed SAR profile and the theoretical model.

IV. Temporal and Spatial Variability of ISWs

This section presents a comprehensive analysis of the temporal and spatial variations of ISWs in the Bay of Biscay. The analysis begins with the extraction of ISW characteristics from SAR images, followed by an examination of their temporal and spatial variation patterns. Next, a specific ISW profile and its corresponding sub-image block are used to illustrate the inversion process of dynamic parameters, including wavelength, wave direction, and amplitude. The accuracy of the ISW amplitude inversion method proposed in this article is then validated through a detailed comparison of theoretical and observed profiles. The phase velocity is estimated by tracking the movement of ISW features across successive images. Finally, the relationship between the ISW generation time and tidal currents is explored based on the inverted ISW phase velocity. In addition, by analyzing the variations in ISW wavelength, wave direction, and amplitude, the structural evolution and intensity changes of ISWs during propagation are thoroughly investigated.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

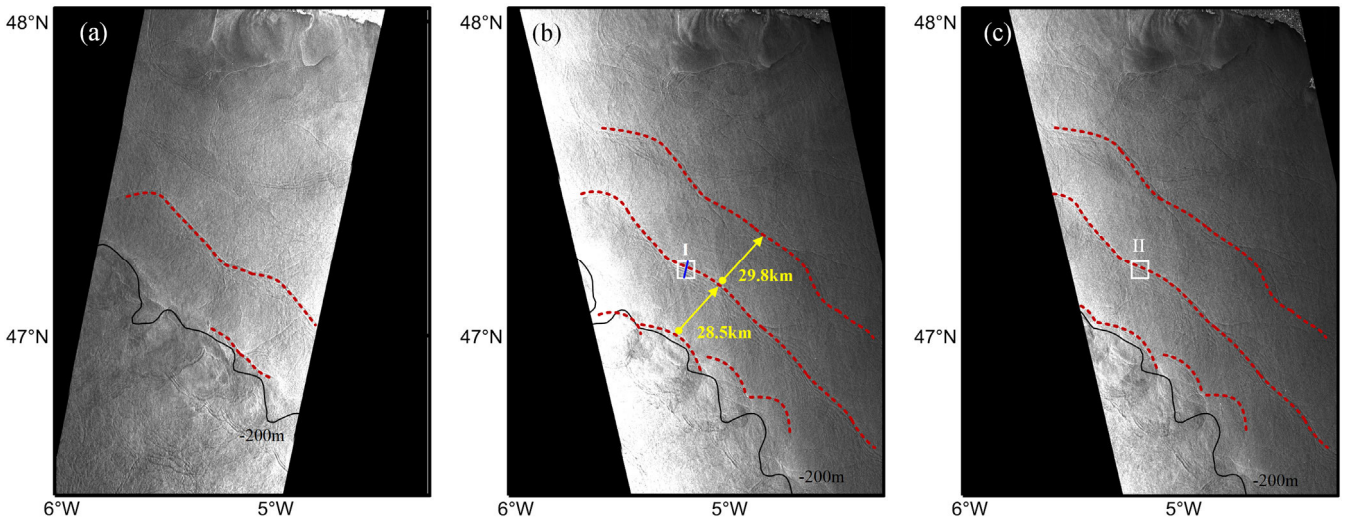


Fig. 3. ENVISAT SAR images acquired at 10:37:54 UTC (a) and 21:57:21 UTC (b) on June 5, 2008. (c) ERS-2 SAR image acquired at 22:27:38 UTC on June 5, 2008. The black curve represents the −200 m depth contour. The red dashed line indicates the position of the ISWs. The white boxes (Windows I and II) denote the selected regions used to invert the wavelength and direction of the ISWs, as discussed in Section IV-B. The blue line indicates the intercepted ISW profile used to invert the ISW amplitude. The yellow arrows show the distance between the ISWs.

A. Evolution of Surface Features in Time-Series SAR Images

The time-series SAR images analyzed in this study include ENVISAT and ERS-2 data, downloaded from the ESA website. These images were radiometrically calibrated and precisely registered to ensure accurate analysis. To optimize computational efficiency, the images were processed with a five-look averaging in both azimuth and range directions, increasing the pixel interval from 12.5 to 62.5 m while maintaining sufficient resolution for analysis. The processed SAR images were mosaicked to cover the entire study area, and the region of interest corresponding to the ISWs was extracted, as shown in Fig. 3.

Fig. 3 presents the SAR images acquired on June 5, 2008, and their corresponding ISW characteristics. Fig. 3(a) is an ENVISAT SAR image acquired at 10:37:54 UTC, showing a group of ISWs roughly parallel to the −200 m depth contour. The curvature of the ISWs suggests they are propagating toward the continental shelf. Multiple ISW groups are observed near the −200 m contour, with their surface characteristics closely resembling those of the previously observed ISWs, indicating a common generation mechanism.

Fig. 3(b) is an ENVISAT SAR image acquired at 21:57:21 UTC, about 11 h and 20 min after Fig. 3(a). This image shows three distinct ISW groups, each approximately parallel to the depth contour. Near the −200 m isobath, several ISW packets, each containing 3–6 waves, are observed. The curvature of these packets follows the isobath, suggesting they may be generated by topographic ridge radiation between the canyons [4]. As the ISWs propagate, they gradually intersect and merge, forming a continuous wavefront toward the continental shelf. Previous studies have indicated that the distance between ISWs generated by tidal disturbances is approximately 28.6 km [1], with the spacing between the groups in Fig. 3(b) being 28.5 and 29.8 km, consistent with mooring data measurements in previous studies.

Fig. 3(c) is an ERS-2 SAR image acquired at 22:27:38 UTC, about 30 min after Fig. 3(b). The ISW features in both images are highly similar, showing strong correlation, which provides a basis for extracting the ISW phase velocity by tracking the motion of ISW features between the two images.

B. Case Studies of Dynamical Parameter Retrieval

This section presents case studies using the ISW profile [blue line in Fig. 3(b)] and its corresponding sub-image block [white window in Fig. 3(b)] to demonstrate the inversion and verification of ISW dynamic parameters, including wavelength, wave direction, and amplitude. The results for extracting the ISW phase velocity from time-series SAR images are also analyzed in detail.

1) Results of Wavelength and Wave Direction Estimation: Fig. 3(b) and (c) show that ISWs appear as distinct bright and dark stripes in the SAR images. To analyze their characteristics, two sub-image blocks (I and II), indicated by the white boxes in Fig. 3, were selected and processed using a 2-D fast Fourier transform (2-D FFT) to obtain the spatial spectrum of ISW energy, as shown in Fig. 4. The spectrum contains multiple peaks due to the presence of both ocean waves and ISWs. In sub-image I, the ocean wave has a wavelength of approximately 320 m, while the ISW wavelength is around 1141 m. In sub-image II, the ocean wave wavelength is approximately 342 m, and the ISW wavelength is about 1313 m.

Previous studies by Pingree and Mardell [1], based on mooring data, suggested that the ISW wavelength in this region is approximately 1 km. New and Silva [14], based on ERS SAR image analysis, reported that the average wavelength of approximately 25 ISW groups in this region is around 1.35 km [14], which aligns well with the SAR-based inversion results in this study. The image spectrum reveals a 180° ambiguity in the ISW wave direction. However,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

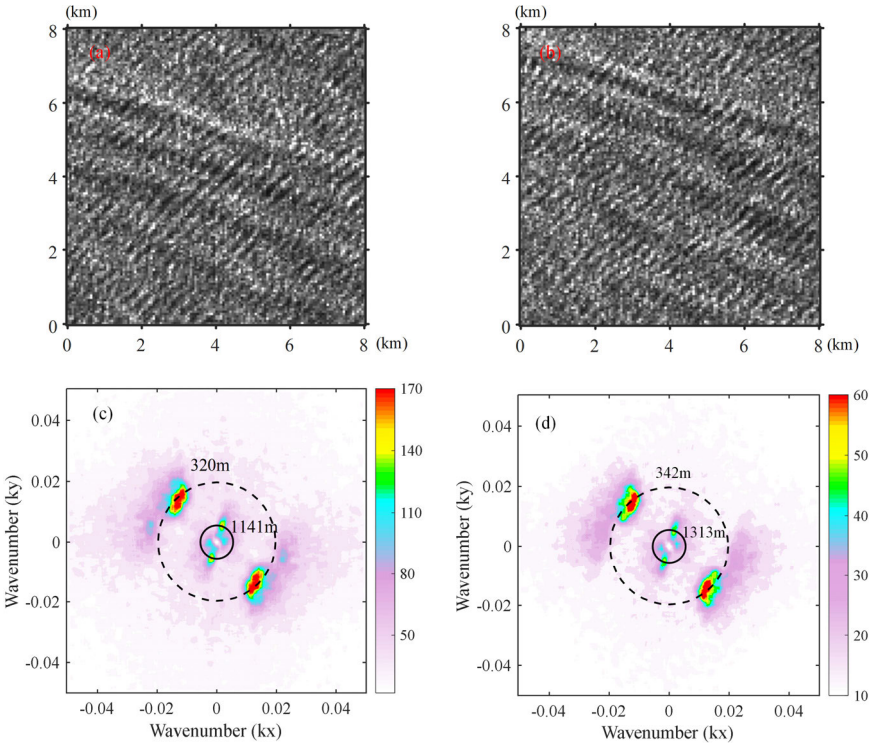


Fig. 4. (a) Sub-image block I and (b) sub-image block II with their corresponding 2-D wavenumber spectra (c) and (d).

by incorporating motion information from the time-series SAR images (as detailed in the phase velocity extraction results), it was determined that the ISW propagates from window I to window II. Consequently, the ISW wave direction in window I [Fig. 4(a)] is 49°, and in window II [Fig. 4(b)] it is 42°.

2) Results of Phase Velocity Extraction: To minimize the influence of ocean wave features on the SAR image, wave textures were initially removed using frequency-domain filter. Based on the typical wavelengths of ocean waves and ISWs calculated in Section IV-B, Item 1, the cutoff wavenumber for the frequency filter is set to 0.01 m^{-1}. After filtering, an image pair was then created by combining the ENVISAT SAR image (10:37:54 UTC, June 5, 2008) and the ERS-2 SAR image (22:27:38 UTC, June 5, 2008). The ISW phase velocity was automatically extracted using the maximum cross correlation method with a template size of 12 km and an overlap of 50%. The correlation coefficient and reciprocal filters were used in the raw velocity filed to eliminate spurious vectors. The correlation coefficient filter threshold was 0.3.

Fig. 5(a) shows a pseudo-color composite of the time-series SAR images. The red component represents the ENVISAT SAR image, and the green component represents the ERS-2 SAR image. The composite clearly illustrates that the ISWs have moved from the red to the green position over approximately 30 min, maintaining strong correlation throughout. This movement allows for the determination of ISW phase velocity by tracking ISW features in the time-series data.

The phase velocity results, presented in Fig. 5(b), reveal that 3–4 groups of ISWs are generated by the interaction between tidal forces and continental shelf fracture topography. These ISWs propagate in a southwest-northeast direction with a consistent speed of approximately 0.3–0.8 m/s. Additionally, several ISWs propagate in the northeast-southwest and northwest-southeast directions, exhibiting phase velocities greater than 1 m/s. The phase velocity results align well with the observed ISW movement in the pseudo-color composite image, validating the extraction accuracy.

Fig. 6 shows the ISW phase velocity superimposed on the regional water depth distribution. The red curve traces the ISWs generated by the interaction between tidal forces and the continental shelf fracture terrain. In the early stages, the crest of the ISW shows a convex shape, coinciding with the convexity of the isobath, suggesting that the ISWs may radiate from terrain ridges between the canyons [4]. As the ISWs propagate, they gradually merge into a continuous wavefront, and the crest line becomes increasingly parallel to the isobath as it moves toward the continental shelf.

Fig. 7 presents the statistical analysis of the phase velocity and propagation direction of the target ISWs. Fig. 7(a) and (b) show the statistical results for phase velocity and propagation direction, respectively. The average phase velocity is approximately 0.54 m/s, and the average propagation direction is about 84°. In comparison, Pingree and Mardell [1] found the ISW phase velocity to be around 0.69 m/s based on mooring data, while New and Pingree [2] derived a value of approximately 0.6 m/s using ADCP data. It is important to note that the phase velocity extracted from the time-series SAR images represents an average over the time interval between acquisitions, whereas the values reported in previous studies correspond to instantaneous measurements. Thus, some discrepancy between these measurements is expected.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

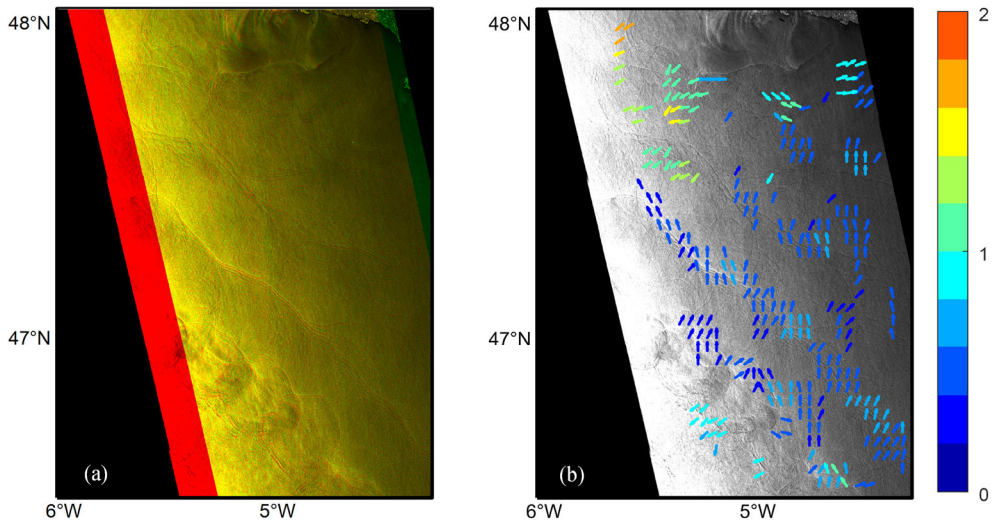


Fig. 5. (a) Pseudo-color composite of the ENVISAT SAR image acquired at 10:37:54 UTC (red component) and the ERS-2 SAR image acquired at 22:27:38 UTC (green component) on June 5, 2008. (b) Internal wave phase velocity extracted by tracking ISW features in time-series SAR images.

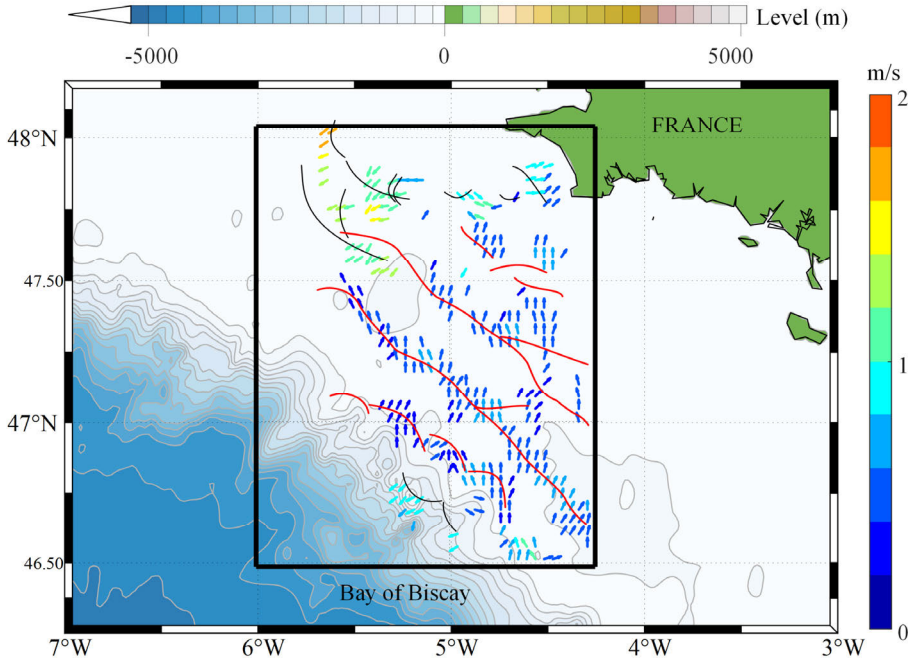


Fig. 6. Superimposed ISW position and phase velocity data on the regional water depth distribution in the study area.

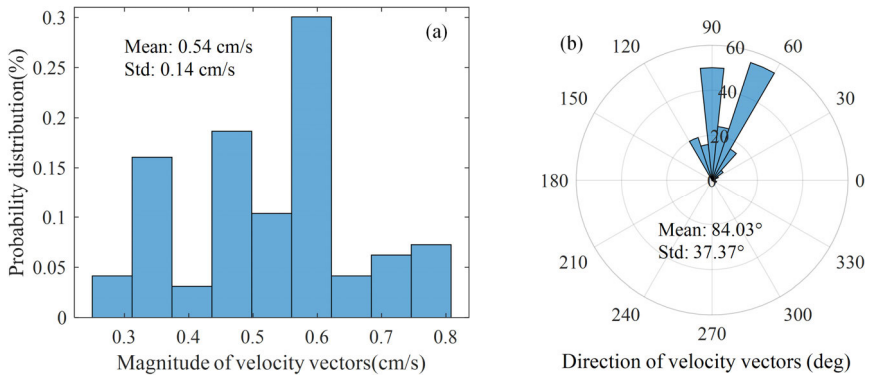


Fig. 7. Statistical analysis of the (a) phase velocity and (b) propagation direction of the target ISWs in the study area.

3) Results of Amplitude Inversion: To retrieve the ISW amplitude, a profile perpendicular to the ISW crest from the ENVISAT SAR image acquired at 21:57:21 UTC on June 5, 2008 [indicated by the blue profile in Fig. 3(b)] was selected

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

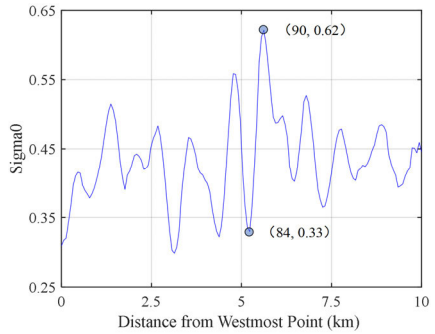


Fig. 8. Radar backscatter coefficient along the blue ISW profile in Fig. 3(b).

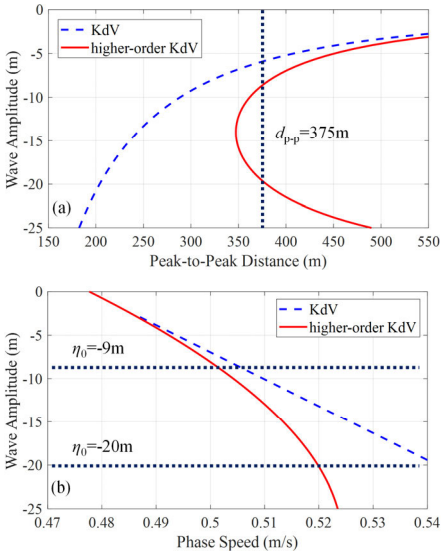


Fig. 9. Variation of internal wave amplitude with (a) peak-to-peak distance and (b) ISW phase velocity at the red internal wave profile in Fig. 3(b).

to calculate the peak-to-peak distance. The radar backscatter coefficient distribution along this profile is shown in Fig. 8, with the positive and negative peak positions and their corresponding backscatter coefficients marked. The peak-to-peak distance of the ISW is approximately 375 m, calculated by multiplying the position difference between the peaks by the SAR image pixel interval. Since the relationship between the peak-to-peak distance and wavelength is not a fixed 2:1 ratio, the peak-to-peak distance here is measured directly from the SAR images, rather than derived from the wavelength.

To estimate the possible range of ISW amplitude, this study analyzes the relationship between ISW amplitude, peak-to-peak distance, and phase velocity. The total water depth along the ISW profile is approximately 153 m, based on the global multi-resolution topography (GMRT) data (https://www. gmrt.org/) from the northern Bay of Biscay. The upper layer depth in this region is approximately 48 m, with a density ratio of $\Delta\rho/\rho_0 = 7.4 \times 10^{-4}$ [5], and the lower layer depth is around 105 m. Substituting these parameters into (16) yields the ISW amplitude variation with peak-to-peak distance under the high-order KdV equation, shown by the red curve in Fig. 9(a). For a peak-to-peak distance of 375 m, the two possible ISW amplitude solutions are −9 and −20 m.

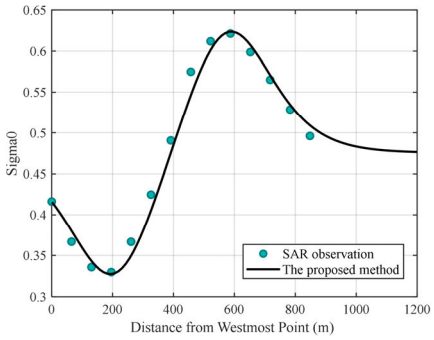


Fig. 10. Comparison of observed SAR profile and theoretical profiles of ISW.

Furthermore, substituting the hydrological data into (9), the ISW amplitude variation with phase velocity is obtained, as shown in Fig. 9(b). Based on Fig. 5(b), the ISW phase velocity is approximately 0.52 m/s, which corresponds to an ISW amplitude of approximately −20 m in Fig. 9(b). Since this solution aligns with the second value in the peak-to-peak distance curve, the possible ISW amplitude range is defined as [−18, −22 m].

To determine the optimal ISW amplitude and upper layer depth, this study defines the upper layer depth range as [38, 58 m], with the estimated depth of 48 m as the center. A series of calculations were performed across the possible ISW amplitude range and upper layer depth interval. The optimal fitting result was obtained by minimizing the mean square error between the observed SAR profile and the theoretical profile. The minimum error of approximately 1.76×10^{-4} occurs when the ISW amplitude is −19.6 m and the upper layer depth is 44.28 m, providing the best fit between the SAR image and theoretical profile.

As there are no in situ measurements for verifying the amplitude inversion accuracy, the inverted ISW amplitude was substituted into the radar backscatter coefficient theoretical formula to calculate the theoretical SAR profile. This profile was compared with the observed one, as shown in Fig. 10. The close agreement between the theoretical and observed profiles confirms the high accuracy of the proposed method.

C. Generation Timing and Tidal Correlation

This section uses location data of ISWs and their phase velocity, extracted from time-series SAR images, to estimate their generation times. The distribution of the ISWs of interest in the study area is shown in Fig. 11.

Fig. 11(a) and (b) display ENVISAT SAR images acquired at 10:37:54 UTC and 21:57:21 UTC on June 5, 2008, respectively, separated by 11 h and 20 min. As observed in Fig. 11(a), ISW a has propagated toward the continental shelf, while ISW b is in the generation stage. Based on the acquisition time of the ENVISAT SAR image in Fig. 11(a), the generation time of ISW b is estimated to be between 09:30 and 11:30 UTC on June 5. Given that previous studies indicate ISWs in this region have a tidal period of approximately 12.5 h [5], the generation time of ISW a is therefore estimated to be between 21:00 and 23:00 UTC on June 4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

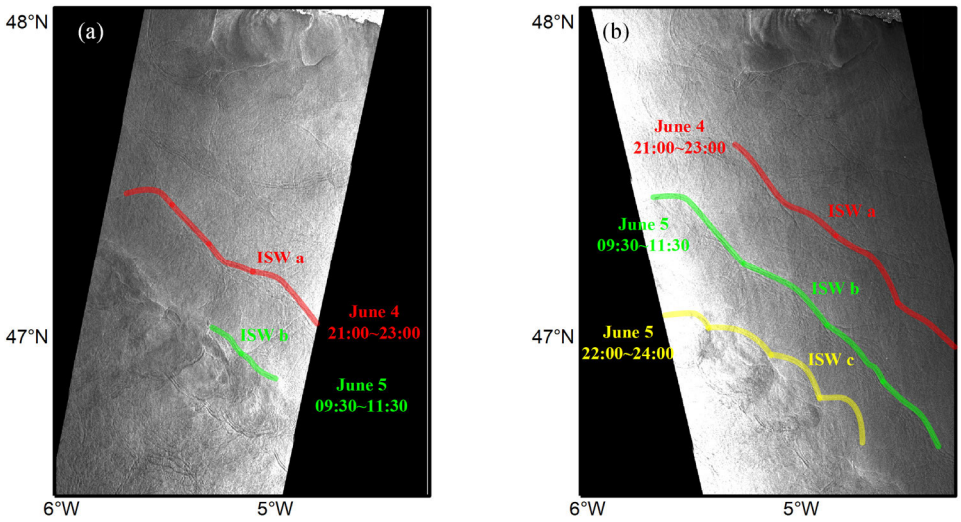


Fig. 11. ENVISAT time-series SAR images acquired at 10:37:54 (a) and 21:57:21 (b) UTC on June 5, 2008. The estimated generation times of the ISWs are marked in the figure.

The ISW phase velocity, extracted from the time-series SAR images, is used to predict the position of ISW in Fig. 11(b). Assuming a phase velocity of 0.54 m/s, ISW propagates approximately 22 km during the 11-h and 20-min interval. This northeastward displacement of ISW a aligns with the red curve in Fig. 11(b), confirming that the red curve corresponds to the same group of ISWs as ISW a in Fig. 11(a). Similarly, ISW b in Fig. 11(a) propagates to the position indicated by the green curve in Fig. 11(b), with its generation time estimated to be between 09:30 and 11:30 on June 5. By subtracting 12.5 h from the generation time of ISW b, the generation time of ISW c is estimated to be between 22:00 and 24:00 on June 5.

While the ISW phase velocity is influenced by factors such as water depth, stratification, and tidal currents, leading to potential errors in estimating propagation distances and generation times, further analysis indicates that these errors have a negligible impact on the correlation between ISW generation and tidal currents.

To explore the relationship between ISW generation times and tidal currents in the northern Bay of Biscay, physical ocean model reanalysis tidal data for the northern Bay of Biscay were obtained from the Copernicus website (https://data.marine.copernicus.eu). The product ID is IBI_MULTI YEAR_PHY_005_002, with a spatial resolution of 0.083° and a temporal resolution of 1 h. The tidal data corresponding to the SAR image acquisition times, both temporally and spatially, were selected, as illustrated in Fig. 12.

The tidal data for the periods of the two SAR images show significant similarities in both space and time. Specifically, the tidal current direction is offshore, and the current intensity is higher at the continental shelf break at the same latitude. This suggests a strong interaction between the tidal current and the shelf break topography, which likely plays a critical role in the generation of ISWs.

Since the ISWs in the study area are generated by the interaction between tidal currents and the continental shelf topography, the region with significant tidal-topography interaction (indicated by the red dashed box in Fig. 12) was selected for statistical analysis. The tidal current intensity in this area was then examined.

Fig. 13 presents the variation curves of the eastward (red) and northward (blue) components of the tidal current from June 4–5, 2008. The yellow bars mark the generation times of ISWs a, ISW b, and ISW c, while the green solid line represents the acquisition times of the time-series SAR images. The figure reveals that all ISWs in the study area occurred during southwest tidal currents (offshore), when the interaction between the tidal currents and the continental shelf topography generated ISWs, which then propagated toward the continental shelf.

Fig. 14 shows the variation of total tidal current intensity over the same period, with the yellow bars indicating the ISW generation times. The results demonstrate that ISWs typically from 1 to 2 h after the peak offshore tidal current. This finding aligns with the analysis by Pingree and Mardell [1] and Pingree et al. [5], who reported that ISWs are generated approximately 1 h after the peak offshore current, based on ADCP and mooring measurements, further confirming the results observed in this study using time-series SAR images.

D. Dynamical Variations During ISW Propagation

After their generation at the continental shelf break in the northern Bay of Biscay, the ISWs propagate toward the continental shelf. Key dynamic parameters, including wavelength, wave direction, phase velocity, and amplitude, are essential for describing the spatiotemporal evolution of ISWs during their propagation.

Fig. 15 shows an ENVISAT SAR image, acquired at 21:57:21 UTC on June 5, 2008, which reveals three distinct groups of ISWs with varying characteristics. These ISWs, generated at the shelf break, move toward shelf. To analyze their dynamic changes, several sub-image blocks (blue boxes in Fig. 15) were selected from these groups based on different propagation distances to calculate wavelength and wave direction. Additionally, profiles (blue lines in Fig. 15) were

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

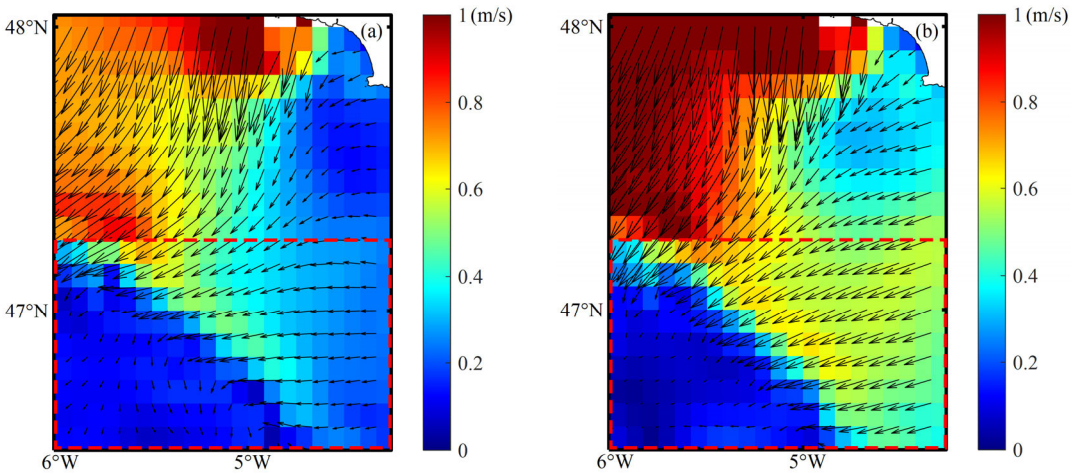


Fig. 12. Tidal data from the Bay of Biscay, temporally and spatially matched with the time-series SAR images. The tidal data in (a) and (b) were acquired at 10:00 and 22:00 UTC on June 5, 2008, respectively. The red dashed box highlights the tidal statistical area selected for further analysis.

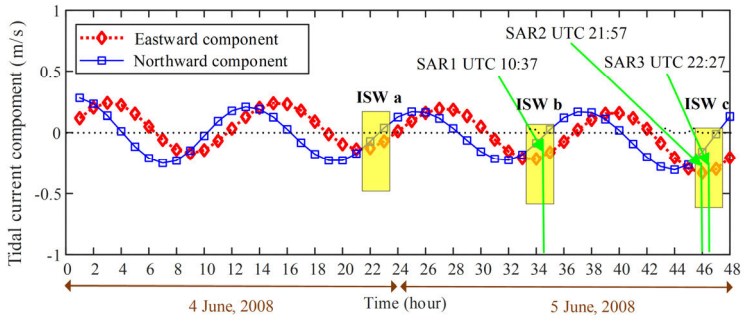


Fig. 13. Variation curve of tidal current components from June 4 to 5, 2008.

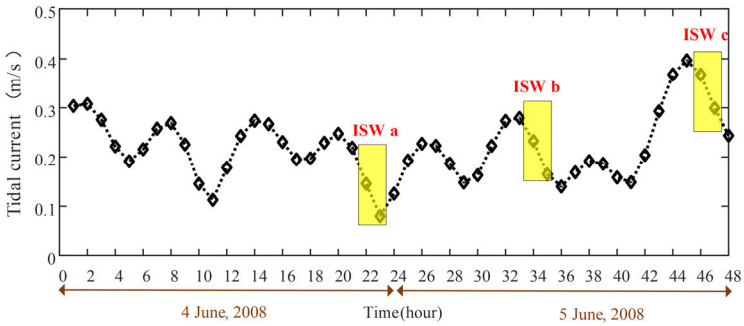


Fig. 14. Variation of total tidal current intensity from June 4 to 5, 2008.

selected for ISW amplitude inversion. Sub-image block 5 and its profile correspond to the white box and profile in Fig. 3(b).

Using the method outlined in Section III, the wavelength and wave direction for each sub-image block were calculated. Amplitude inversion was performed using the motion-information-constrained method proposed in this study. The results, summarized in Table II, show that in the initial stage, the ISW wavelength is approximately 600 m, the peak-to-peak distance is about 300 m, with an amplitude of about 22 m. As the ISWs propagate, both the wavelength and peak-to-peak distance gradually increase, while the amplitude decreases. At approximately 33 km from the −200 m isobath, the light and dark features of the ISWs are clearly visible. At this distance, the wavelength is around 1 km, the peak-to-peak distance is around 400 m, and the amplitude is approximately 19 m. Overall, the ISWs propagate in a southwest-northeast direction, with energy gradually dissipating as they move farther from the source.

The wavelength results in this study are consistent with previous research. For example, New and Pingree [7] found that ISW wavelengths at the Biscay Bay continental shelf break range from 1.0 to 2.0 km, aligning with the wavelengths for ISW a and ISW b in this study. Due to sampling limitations, wavelengths smaller than 1 km cannot be verified [1], so the wavelength for ISW c cannot be confirmed. However, the observed increase in wavelength (from 600 m to 2 km) is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

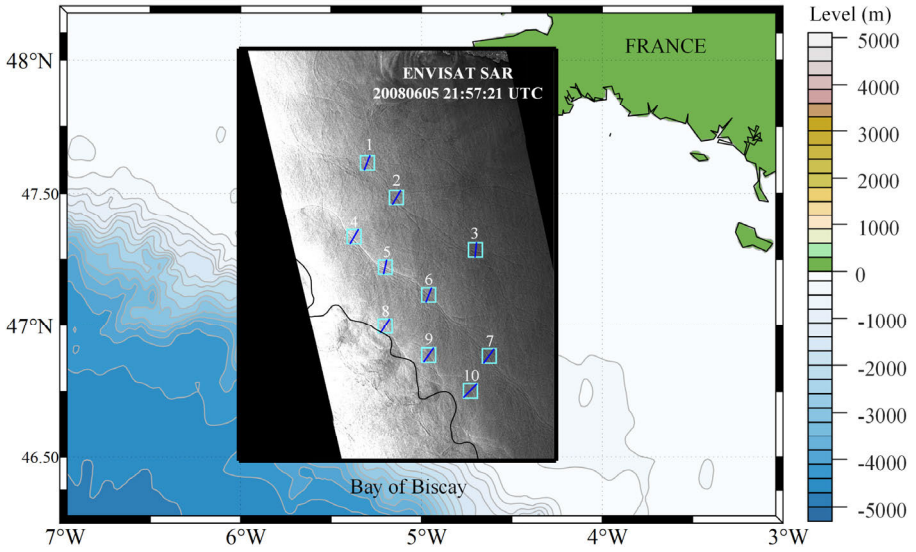


Fig. 15. Distribution of sub-image blocks and profiles selected for the inversion of ISW dynamic parameters in the Bay of Biscay.

TABLE II
INVERSION RESULTS OF DYNAMIC PARAMETERS OF ISWS IN THE BAY OF BISCAY

ISWs	Profile number	Wavelength (m)	Direction(°)	Peak-to-Peak distance (m)	Amplitude(m)	Distance from -200 depth (km)
ISW a	1	1615	48	439	11.6	63.1
	2	1615	42	488	13.5	
	3	1240	49	503	11	
ISW b	4	1306	41	392	17.1	33.3
	5	1141	49	375	19.6	
	6	800	51	361	17.8	
	7	1045	40	425	20.2	
ISW c	8	686	38	336	22	4.8
	9	522	55	350	20.8	
	10	616	37	298	22	

in line with findings from New and Silva [14], who noted that ISW wavelengths increase with distance from the source. This comparison supports the validity of the motion-constrained method used here and indicates that ISW propagation in the northern Bay of Biscay follows similar trends observed in other regions.

V. CONCLUSION

This study investigates the spatiotemporal variation of ISWs generated at the continental shelf break in the Bay of Biscay. The wavelength and direction of the ISWs were inverted by combining the spatial spectral distribution of SAR images with time-series motion data. The phase velocity was extracted by tracking ISW features across the time-series SAR images, and the amplitude was retrieved using a motion information-based inversion method. The theoretical profile, reconstructed from the inverted amplitude, was validated by comparing it with the actual SAR profile, confirming the effectiveness of the proposed inversion method. The spatiotemporal evolution of ISWs, including their structure and intensity, was analyzed through the dynamic parameters at different life stages.

Results show that ISWs propagate in a southwest-northeast direction, with an average phase velocity of 0.54 m/s. As they propagate, the wavelength and peak-to-peak distance of the ISWs gradually increases, while the amplitude decreases. A clear correlation was found between ISW generation and tidal currents, with ISWs typically forming 1–2 h after the peak tidal current. This observation is consistent with previous studies based on mooring and ADCP data.

This study provides novel methods and insights for understanding the life cycle characteristics of ISWs and their environmental impacts at the continental shelf break in the northern Bay of Biscay. However, the accuracy of the inversion method is limited by the temporal resolution of SAR acquisitions, which can introduce uncertainty when tracking rapidly evolving ISWs. With advancements in remote sensing technology, the acquisition of time-series images will improve, addressing this limitation in the future.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

Future studies could apply the proposed method to other regions with high occurrences of ISWs, driven by strong internal tides. The Amazon region, in particular, presents an opportunity to compare SAR-derived ISW parameters with in situ measurements obtained during the AMAZOMIX 2021 cruise. This comparison would not only validate the method but also expand its applicability to diverse marine environments, contributing to a deeper understanding of ISWs and their dynamic behaviors across different oceanic regions.

ACKNOWLEDGMENT

The authors greatly appreciate the ESA for providing ENVISAT and ERS Synthetic aperture radar (SAR) products and all anonymous reviewers who provided detailed and valuable comments or suggestions to improve this article.

REFERENCES

[1] R. D. Pingree and G. T. Mardell, "Solitary internal waves in the celtic sea," *Prog. Oceanogr.*, vol. 14, pp. 431–441, Jan. 1985.

[2] A. L. New and R. D. Pingree, "Large-amplitude internal soliton packets in the central bay of Biscay," *Deep Sea Res. A. Oceanographic Res. Papers*, vol. 37, no. 3, pp. 513–524, Mar. 1990.

[3] X. H. Xie, Y. Cuypers, P. Bouruet-Aubertot, A. Pichon, A. Lourenço, and B. Ferron, "Generation and propagation of internal tides and solitary waves at the shelf edge of the Bay of Biscay," *J. Geophys. Res., Oceans*, vol. 120, no. 10, pp. 6603–6621, Oct. 2015.

[4] A. L. New, "Internal tidal mixing in the bay of Biscay," *Deep Sea Res. A, Oceanographic Res. Papers*, vol. 35, no. 5, pp. 691–709, May 1988.

[5] R. D. Pingree, G. T. Mardell, and A. L. New, "Propagation of internal tides from the upper slopes of the bay of Biscay," *Nature*, vol. 321, no. 6066, pp. 154–158, May 1986.

[6] S. Muacho, J. C. B. da Silva, V. Brotas, P. B. Oliveira, and J. M. Magalhaes, "Chlorophyll enhancement in the central region of the bay of Biscay as a result of internal tidal wave interaction," *J. Mar. Syst.*, vol. 136, pp. 22–30, Aug. 2014.

[7] A. L. New and R. D. Pingree, "Local generation of internal soliton packets in the central bay of Biscay," *Deep Sea Res. A. Oceanographic Res. Papers*, vol. 39, no. 9, pp. 1521–1534, Sep. 1992.

[8] T. Gerkema, F. A. Lam, and L. R. M. Maas, "Internal tides in the bay of Biscay: Conversion rates and seasonal effects," *Deep Sea Res. II, Topical Stud. Oceanogr.*, vol. 51, nos. 25–26, pp. 2995–3008, Dec. 2004.

[9] X. H. Xie et al., "Large-amplitude internal tides, solitary waves, and turbulence in the central bay of Biscay," *Geophys. Res. Lett.*, vol. 40, no. 11, pp. 2748–2754, Jun. 2013.

[10] A. K. Liu and M. K. Hsu, "Internal wave study in the South China sea using synthetic aperture radar (SAR)," *Int. J. Remote Sens.*, vol. 25, nos. 7–8, pp. 1261–1264, Apr. 2004.

[11] C. Maraldi et al., "NEMO on the shelf: Assessment of the Iberia-Biscay-Ireland configuration," *Ocean Sci.*, vol. 9, no. 4, pp. 745–771, Aug. 2013.

[12] R. M. Asiyabi, A. Ghorbanian, S. N. Tameh, M. Amani, S. Jin, and A. Mohammadzadeh, "Synthetic aperture radar (SAR) for ocean: A review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9106–9138, 2023.

[13] A. Azevedo, J. C. B. da Silva, and A. L. New, "Possible generation sites of internal solitary waves observed by ERS SAR in the central region of the bay of Biscay," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 3, Feb. 2002, pp. 1600–1602.

[14] A. L. New and J. C. B. Da Silva, "Remote-sensing evidence for the local generation of internal soliton packets in the central bay of Biscay," *Deep Sea Res. I, Oceanographic Res. Papers*, vol. 49, no. 5, pp. 915–934, May 2002.

[15] A. Azevedo, J. C. B. da Silva, and A. L. New, "On the generation and propagation of internal solitary waves in the southern bay of Biscay," *Deep Sea Res. I, Oceanographic Res. Papers*, vol. 53, no. 6, pp. 927–941, Jun. 2006.

[16] Chonnaniyah, T. Osawa, A. R. As-Syakur, I. W. G. A. Karang, and J. C. B. da Silva, "On the distinction of seasonal internal solitary waves characteristics in the lombok strait based on multi-satellite data," *Int. J. Remote Sens.*, vol. 45, no. 23, pp. 8742–8757, Aug. 2023.

[17] A. K. Mandal, M. Seemanth, and R. Ratheesh, "Characterization of internal solitary waves in the Andaman sea and Arabian sea using EOS-04 and sentinel observations," *Int. J. Remote Sens.*, vol. 45, no. 4, pp. 1201–1219, Feb. 2024.

[18] K. Fan et al., "Internal wave parameters retrieval from space-borne SAR image," *Frontiers Earth Sci.*, vol. 9, no. 4, pp. 700–708, Aug. 2015.

[19] J. Wang, J. Yang, J. Li, L. Ren, and G. Zheng, "Study on extracting and verifying internal wave parameter of SAR image," *Proc. SPIE*, vol. 9638, Oct. 2015, Art. no. 96380X.

[20] C. M. Tensubam, N. J. Raju, M. K. Dash, and H. Barskar, "Estimation of internal solitary wave propagation speed in the Andaman sea using multi–satellite images," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112123.

[21] G. Cui, Y. Du, X. Xu, and X. Yang, "A robust method for retrieving ocean internal wave parameters from SAR imagery using iceemdan," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 7054–7057.

[22] H. Dong, R. Xu, T. Xiao, and Y. Zhang, "Estimation of internal wave parameters based on SAR images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 7073–7076.

[23] X. Zhang and X. Li, "Satellite data-driven and knowledge-informed machine learning model for estimating global internal solitary wave speed," *Remote Sens. Environ.*, vol. 283, Dec. 2022, Art. no. 113328.

[24] X. Zhang et al., "Oceanic internal wave amplitude retrieval from satellite images based on a data-driven transfer learning model," *Remote Sens. Environ.*, vol. 272, Apr. 2022, Art. no. 112940.

[25] M. Dessert, M. Honnorat, J.-M. Le Caillec, C. Messager, and X. Carton, "Estimating the pycnocline depth from the SAR signature of internal waves in the alboran sea," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9048–9061, 2022.

[26] B. Liu, H. Yang, Z. Zhao, and X. Li, "Internal solitary wave propagation observed by tandem satellites," *Geophys. Res. Lett.*, vol. 41, no. 6, pp. 2077–2085, Mar. 2014.

[27] S. Furtney, R. Romeiser, and H. C. Graber, "Automated retrieval of internal wave phase speed and direction from pairs of SAR images with different look directions," *Remote Sens. Environ.*, vol. 305, May 2024, Art. no. 114084.

[28] A. M. Santos-Ferreira, J. C. B. Da Silva, and J. M. Magalhaes, "SAR mode altimetry observations of internal solitary waves in the tropical ocean—Part 1: Case studies," *Remote Sens.*, vol. 10, no. 4, p. 644, Apr. 2018.

[29] A. M. Santos-Ferreira, J. C. B. da Silva, and M. Srokosz, "SAR-mode altimetry observations of internal solitary waves in the tropical ocean—Part 2: A method of detection," *Remote Sens.*, vol. 11, no. 11, p. 1339, Jun. 2019.

[30] J. M. Magalhaes et al., "Using a tandem flight configuration between Sentinel-6 and Jason-3 to compare SAR and conventional altimeters in sea surface signatures of internal solitary waves," *Remote Sens.*, vol. 15, no. 2, p. 392, Jan. 2023.

[31] Q. Zheng, Y. Yuan, V. Klemas, and X. Yan, "Theoretical expression for an ocean internal soliton synthetic aperture radar image and determination of the soliton characteristic half width," *J. Geophys. Res., Oceans*, vol. 106, no. C12, pp. 31415–31423, Dec. 2001.

[32] J. Pan, D. A. Jay, and P. M. Orton, "Analyses of internal solitary waves generated at the Columbia river plume front using SAR imagery," *J. Geophys. Res., Oceans*, vol. 112, no. C7, Jul. 2007, Art. no. C07014.

[33] J. Xue, H. C. Graber, B. Lund, and R. Romeiser, "Amplitudes estimation of large internal solitary waves in the mid-atlantic bight using synthetic aperture radar and marine X-band radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 6, pp. 3250–3258, Jun. 2013.

[34] R. Romeiser and H. C. Graber, "Advanced remote sensing of internal waves by spaceborne along-track InSAR—A demonstration with TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6735–6751, Dec. 2015.

[35] T. Jia, J. Liang, X.-M. Li, and K. Fan, "Retrieval of internal solitary wave amplitude in shallow water by tandem spaceborne SAR," *Remote Sens.*, vol. 11, no. 14, p. 1706, Jul. 2019.

[36] European Space Agency. (2007). *ASAR Product Handbook*. [Online]. Available: https://earth.esa.int/handbooks/asar/CNTR

[37] European Space Agency. (2016). *ERS-2 Product Handbook*. [Online]. Available: https://sedur.sedoo.fr/ctoh/Product_Handbook/ERS2_CTOH_Product-Handbook_1.1.pdf

[38] S. Si-Min, S. Zhu, K. Yan-Yan, Z. Wenjing, and C. Guang-Song, "Simulation analysis for remote sensing inversion of wavelength and water depth by the fast Fourier transform method," *J. East China Normal Univ.*, vol. 2019, no. 2, p. 184, Mar. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.

[39] K. Fan et al., *Synthetic Aperture Radar Ocean Internal Wave Remote Sensing Detection Technology and Application*. Beijing, China: China Ocean Press, 2017.
[40] W. Alpers, "Theory of radar imaging of internal waves," *Nature*, vol. 314, no. 6008, pp. 245–247, Mar. 1985.
[41] W. J. Emery, A. C. Thomas, M. J. Collins, W. R. Crawford, and D. L. Mackas, "An objective method for computing advective surface velocities from sequential infrared satellite images," *J. Geophys. Res., Oceans*, vol. 91, no. C11, pp. 12865–12878, Nov. 1986.
[42] X. Li, J. Morrison, L. Pietrafesa, and A. Ochadlick, "Analysis of oceanic internal waves from airborne SAR images," *J. Coastal Res.*, vol. 15, no. 4, pp. 884–891, 1999.
[43] I. J. Barton, "Ocean currents from successive satellite images: The reciprocal filtering technique," *J. Atmos. Ocean. Technol.*, vol. 19, no. 10, pp. 1677–1689, Oct. 2002.
[44] T. P. Stanton and L. A. Ostrovsky, "Observations of highly nonlinear internal solitons over the continental shelf," *Geophys. Res. Lett.*, vol. 25, no. 14, pp. 2695–2698, Jul. 1998.



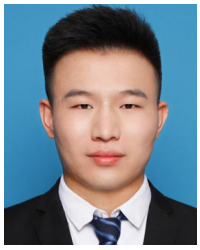
Xue Yang received the B.S. degree in electronic engineering from Dalian Maritime University, Dalian, China, in 2019. She is currently pursuing the Ph.D. degree in signal and information processing with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, and the University of Chinese Academy of Sciences, Beijing.

Her research interests include microwave ocean remote sensing and image information processing.



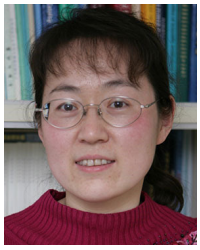
Zongze Li received the B.S. degree in communication engineering from Jilin University, Changchun, China, in 2020. She is currently pursuing the Ph.D. degree in signal and information processing with the Aerospace Information Research Institute, Chinese Academy of Sciences Beijing, China, and the University of Chinese Academy of Sciences, Beijing.

Her research interests include microwave ocean remote sensing and image information processing.



Yawei Zhao received the B.S. degree from China University of Mining and Technology, Xuzhou, China, in 2017, and the Ph.D. degree from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, and the University of Chinese Academy of Sciences, Beijing, in 2022.

He is currently with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include radar signal processing and microwave remote sensing of ocean.



Jinsong Chong (Senior Member, IEEE) received the B.S. degree from Jilin University, Changchun, China, in 1991, the M.S. degree from Beijing University of Aeronautics and Astronautics, Beijing, China, in 2000, and the Ph.D. degree from the Graduate School, Chinese Academy of Sciences, Beijing, in 2003.

She was a Visiting Scholar with Monash University, Melbourne, Australia, in 2004. She was a Tan Chin Tuan Exchange Fellow with Nanyang Technological University, Singapore, in 2005. She is currently a Research Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. Her research interest focuses on microwave remote sensing of ocean.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 15:11:26 UTC from IEEE Xplore. Restrictions apply.